

Stock Market Portfolio Management System

Project Objective

Build a system where users can:

- Create investment portfolios
- Buy and sell stocks
- Track profits/losses
- View portfolio performance
- Simulate stock market trading

This project mimics a simplified version of apps like:

- Zerodha
 - Groww
 - Robinhood Markets
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Core Concepts Used

Concept	Usage
Classes & Objects	Stocks, Portfolio, User
File Handling	Save portfolio data
STL Vectors/Maps	Store stocks
OOP Principles	Encapsulation, inheritance
Exception Handling	Invalid transactions
Searching/Sorting	Stock analysis
Dynamic Memory	Portfolio management

Main Modules

1. User Management

Features

- User registration
 - Login system
 - Password verification
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2. Portfolio Management

Features

- Create portfolio
- Add stocks
- Remove stocks
- View holdings

Example:

Stock	Quantity	Buy Price
AAPL	10	₹170
TSLA	5	₹250

3. Stock Market Module

Features

- Display stock prices
- Simulate market fluctuations
- Update stock values

Example stocks:

- AAPL
 - TSLA
 - GOOGL
 - NVDA
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4. Buy/Sell System

Features

- Buy shares
 - Sell shares
 - Deduct balance
 - Validate transactions
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5. Profit & Loss Analysis

Features

- Total investment
 - Current portfolio value
 - Profit/loss percentage
 - Best performing stock
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6. File Handling System

Store:

- User data
- Portfolio records
- Transaction history